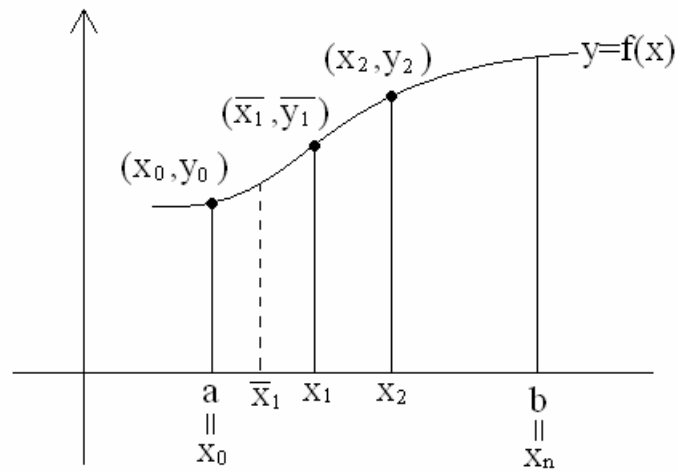


§7-7 Approximate Integration

Homework : 7,19



$$\bar{x}_i = \frac{x_{i-1} + x_i}{2} \quad \Delta x = \frac{b-a}{n}$$

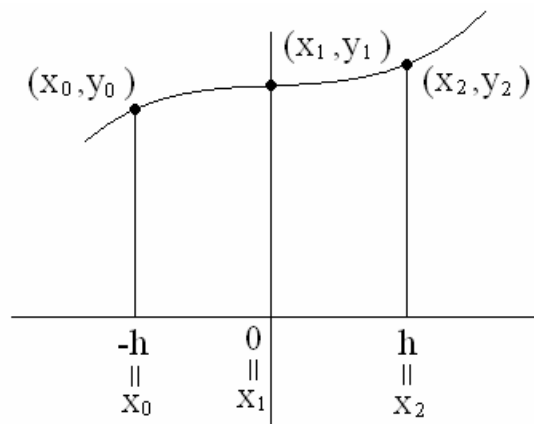
(1) Midpoint Rule : (小長方形的高=小區間中點的 f 值)

$$\int_a^b f(x) \approx \Delta x \left(f(\bar{x}_1) + \dots + f(\bar{x}_n) \right).$$

(2) Trapezoidal : (每小區間用梯形來逼近)

$$\int_a^b f(x) \approx \frac{\Delta x}{2} (y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n).$$

Simpson's Rule : 二小區間的 f 圖上三點用一拋物線來逼近.



找一拋物線 $\Gamma : y = ax^2 + bx + c$ 經過 $(-h, y_0), (0, y_1), (h, y_2)$

$$\begin{aligned}
& \left. \begin{aligned} y_0 &= ah^2 - bh + c \\ \Rightarrow y_1 &= c \\ y_2 &= ah^2 + bh + c \end{aligned} \right\} \Rightarrow y_0 + y_2 \Rightarrow y_0 + 4y_1 + y_2 = 2ah^2 + 6c \\
& \Rightarrow \int_{-h}^h (ax^2 + bx + c) dx = 2 \int_0^h (ax^2 + c) dx \\
& \quad = \frac{h}{3} (2ah^2 + 6c) \\
& \quad = \frac{h}{3} (y_0 + y_2 + 4y_1) \\
& \quad = \frac{h}{3} (y_0 + 4y_1 + y_2).
\end{aligned}$$

也即

$$\int_a^b f(x) dx \approx \frac{\Delta x}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + \dots + 4y_{n-1} + 2y_n)$$

n : must be even

How accurate those approximation are ?

$$\begin{aligned}
\text{誤差} = E_M &\leq \frac{k(b-a)^3}{24n^2}, \quad k = \max_{a \leq x \leq b} |f''(x)|. \\
E_T &\leq \frac{k(b-a)^3}{12n^2}. \\
E_S &\leq \frac{\bar{k}(b-a)^5}{180n^4}, \quad \bar{k} = \max_{a \leq x \leq b} |f^{(4)}(x)|
\end{aligned}$$

Example 1 : $\int_1^2 \frac{1}{x} dx$, $n = 5$, midpoint rule.

Solution :

$$\begin{aligned}
\ln 2 = \int_1^2 \frac{1}{x} dx &\approx 0.2 \left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9} \right) \\
&\approx 0.691908 \\
\Rightarrow \ln 2 &\approx 0.693147.
\end{aligned}$$